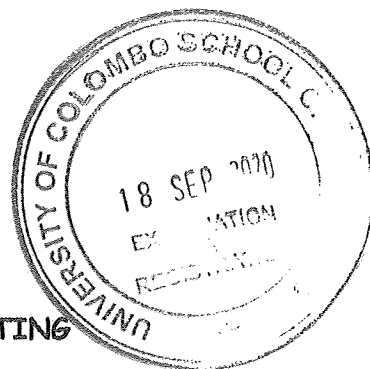




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UNIVERSITY OF COLOMBO, SRI LANKA



UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Academic Year 2019/2020 – First Year Examination – Semester I – 2020

SCS 1204 – Discrete Mathematics I

TWO (2) HOURS

Important Instructions to candidates:

1. The medium of instruction and questions is **English**.
2. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
3. Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
4. Write your index number on each and every page of the answer paper.
5. This paper has **4** questions and **4** pages.
6. Answer **ALL** questions. All questions carry equal marks (**25** marks).
7. Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.
8. Calculators are **not allowed**.

1.

- (a). State the converse, contrapositive, and inverse of the conditional statement "If Nimal is concerned about his cholesterol levels, then he will walk 2 km every day".

[3 Marks]

- (b). Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by forming truth table.

[5 Marks]

- (c). Let $N(x)$ be the predicate "x is a non-negative integer," let $E(x)$ be the predicate "x is even," let $O(x)$ be the predicate "x is odd," and let $P(x)$ be the predicate "x is prime," where the domain for x is the set of integers. Express each of the following statements in terms of $N(x)$, $E(x)$, $O(x)$, $P(x)$, quantifiers, and logical connectives.

- (i). There exists an odd integer.
- (ii). Every integer is odd or even.
- (iii). Not all primes are odd.
- (iv). If an integer is not odd, then it is even.
- (v). The only even prime is 2.

[5 Marks]

- (d). Determine the truth value of each of the following statements if the domain of each variable consists of all real numbers.

- (i). $\exists x \exists y (x + y \neq y + x)$
- (ii). $\forall x \exists y [x + y = 1]$
- (iii). $\exists x \exists y [x + 2y = 7 \wedge 2x + 4y = 10]$
- (iv). $\forall x \exists y [x = y^2]$.

[4 Marks]

- (e). Find the argument form of the following argument and determine whether it is valid.

If Nimal does not study hard, then he is not going to pass the exam.

Nimal passed the exam.

Therefore, Nimal studied hard.

[5 Marks]

- (f). Express the negation of the statement $\forall x \exists y (xy = 1)$ so that no negation precedes a quantifier. Assume that the domain of both x and y is the set of real numbers.

[3 Marks]

2.

- (a). Write down the argument form of the following argument in predicate logic and determine whether it is valid.

Some Computer scientists are Mathematicians.

Some Mathematicians are Engineers.

Therefore, Some Computer scientists are Engineers.

[6 Marks]

- (b). Prove that for any two real numbers x and y , $|xy| = |x||y|$, where $|x|$ is defined as

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

[6 Marks]

- (c). Using Mathematical induction principle, prove that $7^{n+2} + 8^{2n+1}$ is divisible by 57 for every nonnegative integer n .

[7 Marks]

- (d). Let p, q be two integers. Prove that if pq is even then p is even or q is even.

[6 Marks]

3.

- (a). Let A , and B be two subsets of the universal set U , where

$U = \{\text{Apple, bat, cat, dog, egg, fan, goat, hat}\};$

$A = \{\text{bat, dog, egg, hat}\};$ and

$B = \{\text{dog, cat, hat, goat, fan}\}.$

With the usual notations, find the following sets:

- (i). $A \cup B;$
- (ii). $A \cap B;$
- (iii). $\bar{A}$ (or A^c);
- (iv). $A - B;$ and
- (v). $A \oplus B.$

[5 Marks]

- (b). Let A, B , and C be three sets. Prove the following set identities either by showing each side is a subset of the other side or by using a membership table.

- (i). $\overline{(A \cap B)} = \bar{A} \cup \bar{B};$
- (ii). $\overline{(A \cap B \cap C)} = \bar{A} \cup \bar{B} \cup \bar{C};$ and
- (iii). $(A - B) \cap (C - B) = (A \cap C) - B.$

[12 Marks]

(c). Let f be a function from a set A to a set B , and let S and T be two subsets of A .

(i). Show with the usual notations that $f(S \cap T) \subseteq f(S) \cap f(T)$.

(ii). What is meant by f is one-to-one?

(iii). Show that if f is one-to-one then $f(S) \cap f(T) \subseteq f(S \cap T)$.

[8 Marks]

4.

(a). Let $R = \{ (1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4) \}$ be a binary relation on the set $\{1, 2, 3, 4\}$. Check whether R is reflexive, symmetric, anti-symmetric, and transitive. Justify your answer.

[8 Marks]

(b). Let m be an integer such that $m > 1$. Let R be the relation on the set of integers ($\mathbb{Z}$) such that $aRb \Leftrightarrow m \mid (a - b)$. Show that R is an equivalence relation on $\mathbb{Z}$.

[Note that $m \mid (a - b)$ means m divides $(a - b)$]

[8 Marks]

(c). Suppose that $A = \{3, 5, 9, 15, 24, 45\}$ and R is the partial order relation on A defined by $(x, y) \in R$ if and only if $x \mid y$. Note that $x \mid y$ means x divides y .

(i). Draw the Hasse diagram for the relation R .

(ii). Find all minimal elements of A .

(iii). Find all maximal elements of A .

(iv). Find the least element of A (if exists).

(v). Find the greatest element of A (if exists).

[9 Marks]